SIMATS ENGINEERING
ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)
Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A
Research Review & Analysis

Code of Conduct:
*I **RASHMITHA SENTHIL KUMAR (192525034)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the student: **RASHMITHA. S**

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	**Decision Tree Learning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	**Neural Networks vs Statistical Learning:** Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	**Kernel Methods:** Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	**Reinforcement Learning:** Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	**Explainable and Generalizable AI:** Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	**2. Research Problem and Objectives**	Clearly state the research problem and specific objectives of the review.
6	**3. Literature Review**	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	**4. Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	**5. Critical Analysis**	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	**6. Research Gap**	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	**7. Proposed Research Direction**	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	**8. Discussion**	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	**9. Conclusion**	Summarize the major findings, key insights, research gap, and future direction.
13	**References**	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

FROM Q-LEARNING TO DEEP Q-NETWORKS: A COMPARATIVE REVIEW OF REINFORCEMENT LEARNING TECHNIQUES FOR INTELLIGENT DECISION-MAKING

CO4   ASSESSMENT TOOL -2

Submitted in the partial fulfilment for the Course of

MLA0104-Artificial Intelligence and Expert System

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BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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FROM Q-LEARNING TO DEEP Q-NETWORKS: A COMPARATIVE REVIEW OF REINFORCEMENT LEARNING TECHNIQUES FOR INTELLIGENT DECISION-MAKING

ABSTRACT:

Reinforcement Learning (RL) is a major learning paradigm for intelligent agents that must make a sequence of decisions while interacting with an environment. This research review focuses on three representative value-based methods: Q-Learning, State-Action-Reward-State-Action (SARSA), and Deep Q-Networks (DQN). The review follows a structured comparative approach covering learning mechanism, exploration strategy, convergence, computational requirement, scalability, interpretability, classification of application suitability, and evidence from foundational and recent research. Q-Learning is an off-policy temporal-difference method with a strong tabular convergence result under stated conditions, while SARSA is an on-policy method whose updates reflect the behavior policy actually followed. DQN extends Q-Learning to high-dimensional observations by using a neural network together with experience replay and a target network. Research after DQN has improved data efficiency and stability, but deep RL continues to face important limitations including high sample demand, sensitivity to hyperparameters and random seeds, reproducibility, safe exploration, generalization, and limited interpretability. The review identifies these limitations as a combined research gap and proposes an adaptive, safety-aware and explainable value-learning direction. Because this assessment is a research review rather than a mandatory programming exercise, implementation is presented as a proposed evidence-based validation framework rather than as fabricated experimental results. The proposed framework uses benchmark environments, repeated random seeds, learning curves, success rate, sample efficiency, wall-clock time, and stability measures to compare the algorithms fairly. Overall, the review finds that algorithm choice should depend on environment complexity, safety constraints, computational resources, and the level of generalization required.

Keywords: Reinforcement Learning, Q-Learning, SARSA, Deep Q-Network, Exploration-Exploitation, Intelligent Agents

1. INTRODUCTION:

Reinforcement Learning is a machine-learning framework in which an agent learns a policy through repeated interaction with an environment. At each time step, the agent observes a state, selects an action, receives a reward, and moves to a new state. The central objective is not merely to maximize the immediate reward but to learn behavior that maximizes expected cumulative reward over time. This sequential nature distinguishes RL from ordinary supervised prediction and makes it particularly relevant to autonomous software agents, robotics, adaptive control, scheduling, resource allocation, recommendation, and game-playing systems.

The three methods selected in the assessment question represent an important progression in value-based RL. Q-Learning learns an optimal action-value function independently of the behavior policy used to gather experience. SARSA instead learns the value of the policy that is actually being followed, including its exploratory behavior. DQN retains the Q-Learning objective but replaces a discrete Q-table with a deep neural network, allowing action values to be approximated from complex observations such as images. The 2015 DQN work showed that deep RL could learn control policies directly from high-dimensional Atari inputs, making it a landmark in modern reinforcement learning.

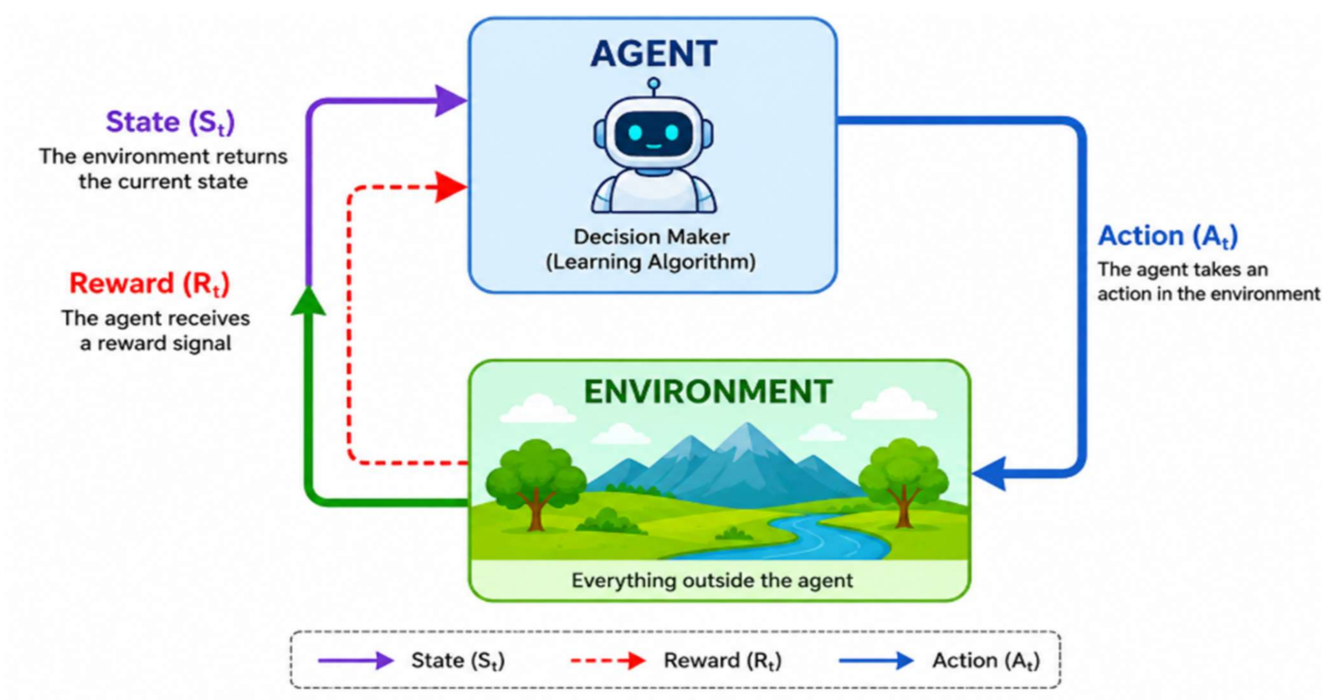

Figure 1: Reinforcement Learning interaction loop between an intelligent agent and its environment.

The motivation for this review is that the three algorithms cannot be judged only by final reward. A method that is computationally inexpensive may fail when the state space becomes very large; a method that learns an aggressive optimal target may behave poorly during exploration; and a deep method that scales to images may require large amounts of data and computation.

Therefore, a technically meaningful review must compare not only performance but also convergence assumptions, exploration behavior, representation capacity, stability, interpretability, computational cost, reproducibility, and suitability for real-world deployment.

The review concentrates on model-free, value-based reinforcement learning. Q-Learning, SARSA, and DQN are analyzed in depth because they are explicitly required by the selected assessment question. Related DQN improvements such as Double DQN and Rainbow are used to explain how later research addressed weaknesses of the original DQN. Policy-gradient and actor-critic families are outside the primary scope and are mentioned only where they help explain broader future directions.

2. RESEARCH PROBLEM AND OBJECTIVES:

A central research problem in reinforcement learning is the selection of a suitable learning method for a particular intelligent-agent task. Q-Learning and SARSA are transparent and computationally light in small discrete environments, but their tabular representation scales poorly as the number of states and actions grows. DQN improves representational scalability by using neural function approximation, but this introduces higher computational cost, instability, sensitivity to implementation choices, and reduced interpretability. The problem addressed by this review is therefore to determine how these three methods differ, what evidence supports their strengths, which limitations remain unresolved, and how future value-based RL could become more efficient, stable, safe, generalizable, and explainable.

1. To review foundational and recent research related to Q-Learning, SARSA, and DQN.
2. To explain the learning mechanism and update rule of each method accurately.
3. To compare exploration, convergence, computational cost, scalability, interpretability, and application suitability.
4. To synthesize findings across multiple studies rather than presenting isolated paper summaries.
5. To critically evaluate methodological limitations, assumptions, reproducibility, and practical deployment challenges.
6. To identify significant research gaps and propose a feasible future research direction.

3. REVIEW METHODOLOGY / ANALYSIS FRAMEWORK:

A structured narrative-comparative review approach was adopted. Foundational sources were retained when they define the algorithms or provide standard theoretical results, while recent peer-reviewed research and surveys were used to identify current limitations and emerging directions. Search concepts included Q-Learning, SARSA, Deep Q-Network, DQN improvements, reinforcement-learning reproducibility, safe reinforcement learning, sample efficiency, robotics, and real-world reinforcement learning. Priority was given to established journals and conferences such as Machine Learning, Nature, AAAI, and recent scholarly surveys.

Sources were included when they directly contributed to at least one required dimension of the assessment: algorithm mechanism, exploration, convergence, computational requirement, real-world application, comparative evidence, research limitation, or future research direction. The analysis uses accuracy of technical claims, strength of evidence, reproducibility, scalability, computational demand, interpretability, and generalization as recurring comparison dimensions. Cross-paper numerical scores are not treated as directly comparable when the studies use different environments, reward scales, preprocessing, or evaluation protocols.

The rubric includes experimental or evidence-based validation. The selected question, however, is explicitly a research review and does not state that software implementation is compulsory. Accordingly, this report uses published experimental evidence as the primary validation source and adds a proposed reproducible implementation framework in Section 9. No experimental values are claimed as results unless they are supported by cited literature.

4. LITERATURE REVIEW AND SYNTHESIS:

Watkins and Dayan established Q-Learning as a model-free, off-policy temporal-difference control method and provided a convergence theorem for the tabular setting under stated sampling and representation conditions [1]. Q-Learning estimates the long-term utility of taking action a in state s and then behaving optimally.

The update target uses the maximum estimated action value at the next state, so the learned target policy can be greedy even while experience is collected using an exploratory behavior policy.

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha\, [r + \gamma\, \max_{a'} Q(s',a') - Q(s,a)]}$$

Its main strengths are simplicity, low computational overhead, model-free learning, and direct interpretability when the state-action table is small. The same table becomes its main limitation as dimensionality grows. A large or continuous state space requires either discretization or function approximation, and naïve discretization can create a very large number of state-action entries. Q-Learning can also exhibit optimistic value estimates because the same noisy estimates are used for both action selection and evaluation, motivating later methods such as Double Q-Learning and Double DQN.

SARSA is an on-policy temporal-difference control algorithm. Its update uses the next action actually chosen by the current behavior policy rather than the maximum action value. This means that the learned action values incorporate the consequences of exploration. Under an epsilon-greedy policy, SARSA therefore evaluates behavior that sometimes takes exploratory actions, while Q-Learning updates toward a greedy target regardless of which exploratory action was actually selected.

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha\, [r + \gamma\, Q(s',a') - Q(s,a)]}$$

The on-policy characteristic can be beneficial in environments where exploratory actions carry risk, because the learned values account for the behavior policy's actual action choices. However, this can also produce a more conservative policy while exploration remains active. Like Q-Learning, tabular SARSA has limited scalability to high-dimensional observations. Its performance is strongly influenced by learning rate, discount factor, reward structure, and the exploration schedule.

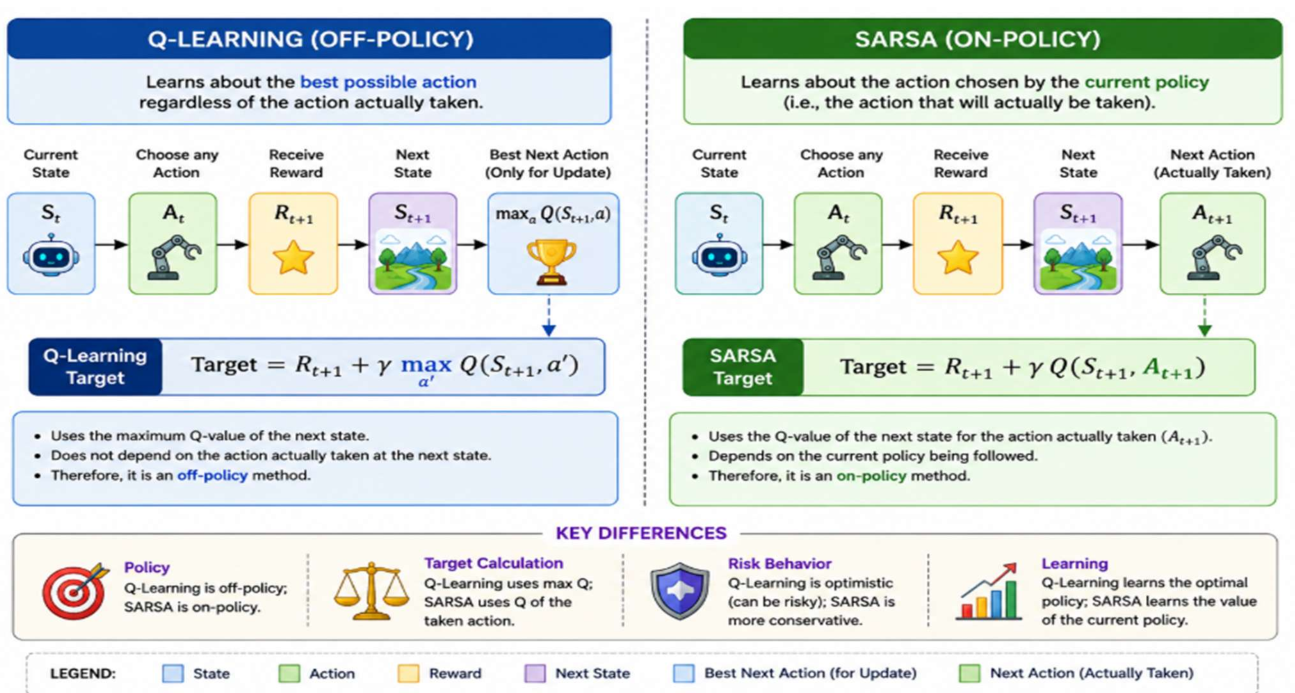

Figure 2: Difference between the Q-Learning off-policy target and the SARSA on-policy target.

Deep Q-Networks extend Q-Learning by approximating the action-value function with a deep neural network. The landmark work by Mnih et al. combined reinforcement learning with deep representation learning and demonstrated learning directly from high-dimensional Atari game inputs [2]. Two engineering ideas were central to the original DQN: experience replay and a separate target network. Experience replay stores transitions and samples them during training, reducing the strong temporal correlation that occurs when consecutive interactions are used directly. The target network is updated more slowly, providing a more stable target for temporal-difference learning.

DQN greatly increases the range of observations that value-based RL can process, but it also introduces new difficulties. Neural function approximation makes optimization non-linear, training can be unstable, performance depends on hyperparameters and random seeds, and large replay buffers and repeated neural-network updates increase memory and compute requirements. Henderson et al. showed that deep-RL comparisons can be difficult to reproduce and interpret when variance and experimental reporting are not handled carefully [3]. Later work such as Rainbow combined multiple DQN improvements and showed that several extensions can complement each other [4].

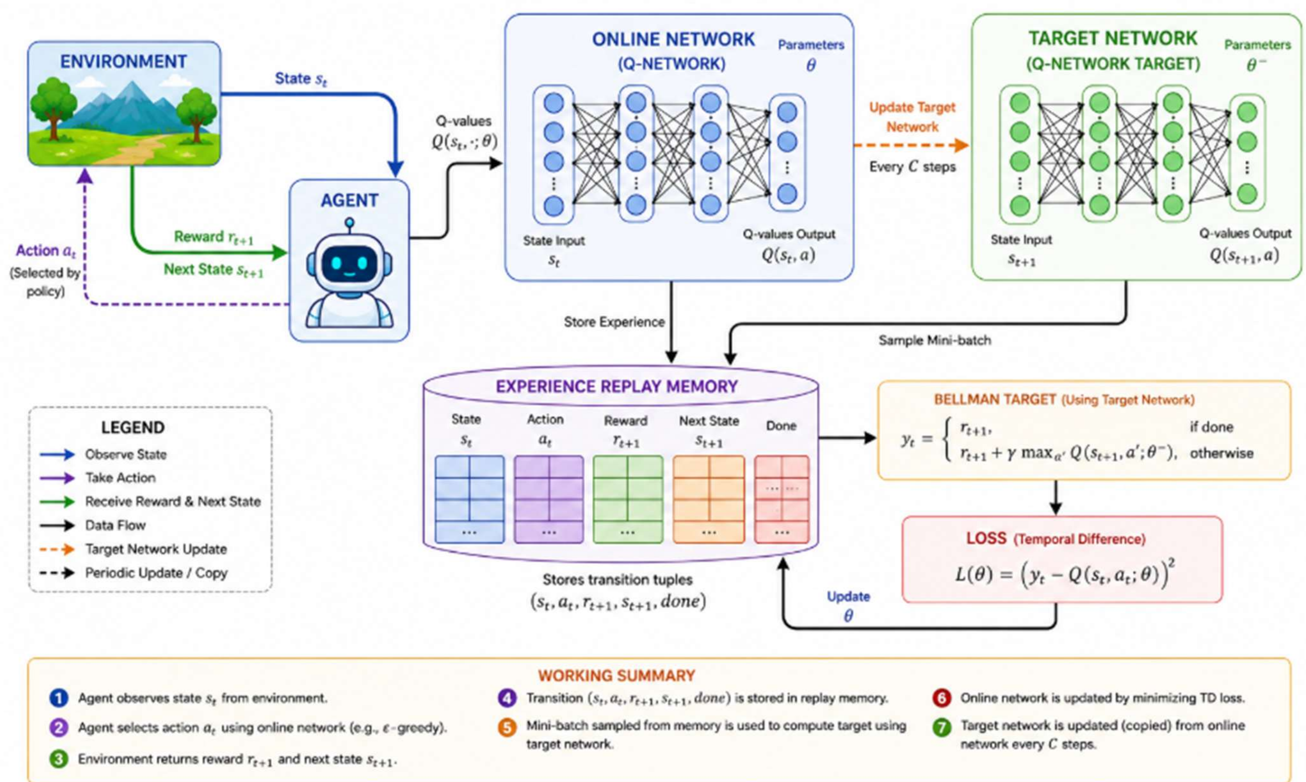

Figure 3: Simplified DQN architecture showing the online network, replay memory, and target network.

The literature reveals a consistent trade-off. Tabular Q-Learning and SARSA are attractive when the environment is small enough to represent explicitly because their updates are inexpensive and their learned values can be inspected directly. Their weakness is representational scalability. DQN reverses this trade-off: it can process high-dimensional observations through learned representations, but it requires substantially more data, memory, computation, and experimental care. The transition from Q-Learning to DQN is therefore not simply an increase in performance; it is a change in the representation and optimization regime, bringing new opportunities and new sources of error.

Table 1: Selected Research Sources and Their Contribution

Source	Method / Focus	Major Contribution	Role in Review
Watkins & Dayan, 1992 [1]	Q-Learning	Formal tabular Q-Learning convergence result under stated conditions.	Foundational theory
Mnih et al., 2015 [2]	DQN	Deep Q-learning with experience replay and target network; Atari evaluation.	Landmark deep RL
Henderson et al., 2018 [3]	Deep RL evaluation	Highlighted variance, reproducibility and reporting problems.	Critical evidence
Hessel et al., 2018 [4]	Rainbow / DQN extensions	Combined six DQN improvements and studied their contributions.	Improvement evidence
Tang et al., 2024 [5]	Deep RL in robotics	Surveyed real-world successes and emphasized stable, sample-efficient real-world RL.	Recent application/gap

5. COMPARATIVE ANALYSIS OF AI TECHNIQUES:

The required comparison is most useful when it separates theoretical properties from practical behavior. Q-Learning and SARSA use the same tabular representation and similar temporal-difference structure, but differ in whether the update target follows a greedy target policy or the behavior policy. DQN is structurally closer to Q-Learning because it is also off-policy value learning, but replaces explicit storage with neural approximation. This increases scalability while reducing direct interpretability and increasing computational demand.

Table 2: Comparative Analysis of Q-Learning, SARSA, and DQN

Parameter	Q-Learning	SARSA	DQN
Policy relationship	Off-policy	On-policy	Off-policy
Representation	Tabular Q-values	Tabular Q-values	Deep neural-network approximation
Update target	Maximum next-state Q-value	Q-value of next action actually selected	Neural approximation of max next-state Q-value
Exploration	Commonly ε-greedy behavior	Commonly ε-greedy current policy	Commonly ε-greedy; training also uses replay
Convergence	Strong result in finite tabular setting under conditions [1]	Tabular convergence under suitable policy/step-size assumptions	No equally general tabular-style guarantee with nonlinear deep approximation
Scalability	Low for very large state spaces	Low for very large state spaces	High for complex/high-dimensional observations
Interpretability	High when table is manageable	High when table is manageable	Lower due to distributed neural representation
Compute / memory	Low / Q-table	Low / Q-table	High / network parameters + replay buffer
Stability	Generally simpler to diagnose	Generally simpler to diagnose	Sensitive to approximation, targets, seeds and hyperparameters
Best fit	Small discrete control tasks	Small discrete tasks where behavior-policy risk matters	Large observation spaces and representation-heavy control tasks

All three methods may use epsilon-greedy exploration, where a random action is selected with probability ε and the currently preferred action is selected otherwise. The effect of exploration differs, however. In Q-Learning, the update still targets the greedy next action, so exploration is separated from the target policy. In SARSA, the exploratory next action enters the update itself. DQN usually inherits epsilon-greedy exploration from Q-Learning, but the replay mechanism allows past transitions to be reused many times. More advanced exploration strategies can improve difficult tasks, but they also add complexity and may change the fairness of algorithm comparisons.

For tabular Q-Learning, the foundational convergence result depends on conditions such as repeated sampling and an appropriate discrete representation [1]. Similar tabular temporal-difference theory supports SARSA under suitable assumptions. Deep Q-learning is more difficult because neural-network approximation, bootstrapping, and off-policy data interact. Experience replay and target networks were introduced to improve practical stability, but they do not create a universal convergence guarantee for arbitrary deep networks. This difference is important: successful empirical training should not be confused with the stronger theoretical guarantees available in simpler tabular settings.

Q-Learning and SARSA require a value for every represented state-action pair, making each update cheap but potentially making storage enormous when the state space is finely discretized. DQN replaces this table with neural-network parameters and can generalize across similar inputs, which is valuable for images and other high-dimensional observations. The cost is repeated forward and backward passes, replay-buffer memory, target-network maintenance, and usually hardware acceleration for larger models. Therefore, DQN's representational scalability does not imply lower computational cost.

6. CRITICAL ANALYSIS OF RESEARCH FINDINGS:

Q-Learning provides a clear and mathematically grounded baseline for off-policy control. Its separation of behavior and target policies is useful when an agent must explore while learning a greedy target. SARSA offers a complementary on-policy perspective and can better reflect the cost of exploratory behavior in the learned value function. DQN's major strength is representation learning: it can map complex observations to action values without manually constructing a table for every state. The literature therefore supports viewing the three algorithms as complementary points on a spectrum rather than as simple replacements for one another.

The tabular algorithms assume that the chosen state representation is sufficiently informative and manageable. If relevant information is omitted, the Markov assumption may be violated; if the representation is too detailed, the table becomes impractical. DQN reduces the need for explicit discretization but assumes that a neural network can learn a useful representation from available experience. It also depends on design choices such as network architecture, replay strategy, target-update frequency, optimizer, reward scale, and exploration schedule. These dependencies make deep-RL results more difficult to reproduce than a simple tabular experiment.

Henderson et al. demonstrated that variance and experimental methodology can make apparent deep-RL improvements difficult to interpret [3]. This has direct implications for the present review: a single training run or a single final reward should not be treated as sufficient evidence. Meaningful validation should use multiple random seeds, learning curves, confidence intervals or dispersion measures, clear hyperparameter reporting, and consistent evaluation conditions. Comparisons should also distinguish sample efficiency from final performance and wall-clock efficiency.

Physical and operational environments add challenges that are often absent from benchmarks. Exploration can damage equipment or create unsafe behavior; collecting interactions can be expensive; environmental dynamics can change; observations may be noisy or partially observable; and learned policies may encounter states not represented during training. A recent robotics survey emphasizes the continuing need for stable and sample-efficient real-world RL as well as principled evaluation [5]. These concerns make safety, generalization, and data efficiency central research requirements rather than optional enhancements.

7. RESEARCH GAP IDENTIFICATION:

The reviewed literature indicates that the most important gap is not a single missing algorithmic trick but the absence of a unified value-learning approach that simultaneously offers the simplicity and transparency of tabular methods, the representational power of deep methods, efficient use of experience, reliable uncertainty handling, safe exploration, and understandable decisions. Existing improvements usually optimize only part of this problem.

- Sample efficiency: deep value-based agents may require large numbers of interactions, which is costly in physical environments.
- Training stability: neural approximation, bootstrapping, off-policy learning, and moving targets can produce unstable learning.

- Value-estimation bias: maximization over noisy estimates can lead to overestimation and distorted action values.
- Safe exploration: ordinary ε-greedy exploration can select unacceptable actions in safety-critical systems.
- Generalization: strong benchmark performance does not guarantee reliable behavior in unseen states, changed dynamics, or new environments.
- Interpretability: DQN action choices are much harder to explain than small Q-tables, limiting trust and debugging.
- Reproducibility: results can vary with random seeds, implementations, preprocessing, and hyperparameters [3].
- Evaluation consistency: different studies frequently use different environments and metrics, making direct numerical comparison difficult.

8. PROPOSED RESEARCH DIRECTION:

A feasible future research direction is an Adaptive Safety-Aware Explainable DQN (ASE-DQN) framework. The objective is not to replace every RL family, but to address the gaps identified in value-based deep RL while retaining a clear connection to Q-Learning. The framework would combine four mechanisms: prioritized or informative experience reuse for sample efficiency, bias-reduced target estimation such as Double-DQN-style selection/evaluation separation, uncertainty-aware or constraint-aware exploration for safety, and an explanation layer that records the state features and value differences that influenced each selected action.

The expected contribution would be a more deployment-oriented value-learning agent. Efficient replay would reduce wasted updates on uninformative transitions; bias-reduced targets would improve value estimation; safety constraints would prevent clearly prohibited exploratory actions; and explanation traces would improve human understanding of why the agent selected a particular action. The research hypothesis would be that these mechanisms can improve stability and safety without sacrificing the main scalability advantage of DQN.

7. Can adaptive replay reduce the number of environment interactions needed to reach a target performance?
8. Can safety-aware exploration reduce constraint violations while preserving learning progress?
9. Can bias-reduced targets improve stability across random seeds?

10. Can lightweight explanations improve interpretability without materially increasing inference latency?
11. Does the combined framework generalize better when environment dynamics are changed after training?

9. PROPOSED IMPLEMENTATION AND EXPERIMENTAL VALIDATION:

The assessment sheet asks for a research review and analysis and does not explicitly require a coded implementation. However, the rubric allocates marks to research methodology and experimental/evidence-based validation. Therefore, including a proposed implementation and validation plan strengthens the report. The plan below is intentionally presented as proposed work; it does not claim that experiments have already been executed.

A reproducible comparison can be implemented in Python using a standard reinforcement-learning environment library and a deep-learning framework. Q-Learning and SARSA should first be tested on discrete benchmark tasks where tabular representation is valid. DQN should be tested on tasks with larger or continuous observation representations after suitable preprocessing. To preserve fairness, common reward definitions and evaluation episodes should be used wherever the environment permits.

The baselines should be standard Q-Learning, standard SARSA, standard DQN, and the proposed ASE-DQN. Each configuration should be trained with multiple random seeds. Evaluation should include mean episodic return, success rate, episodes or interactions required to reach a defined threshold, standard deviation across seeds, training wall-clock time, memory use, number of safety violations, and inference latency. For explainability, a simple human-evaluation or consistency measure can be added to assess whether the explanation corresponds to the action-value difference.

12. Define the environment, state representation, action space, reward function, termination conditions, and safety constraints.
13. Fix a reproducible set of random seeds and report all major hyperparameters.
14. Train each baseline under the same interaction budget wherever technically meaningful.
15. Evaluate periodically without exploration to separate training behavior from final policy performance.
16. Plot learning curves with variability across seeds rather than reporting only the best run.

17. Perform ablation studies by removing replay prioritization, safety filtering, bias reduction, or explanation components one at a time.
18. Interpret results using both final performance and efficiency/stability metrics.

10. RESULT ANALYSIS AND INTERPRETATION:

The published evidence supports three broad interpretations. First, Q-Learning has strong value as a theoretical and practical baseline in finite tabular problems because its convergence properties are well characterized under explicit conditions [1]. Second, DQN demonstrates that value-based RL can be extended to high-dimensional sensory input through deep representation learning [2]. Third, later research shows that DQN performance can be improved substantially by combining complementary mechanisms, while experimental variance and reproducibility remain important concerns [3][4].

For small discrete problems with a manageable state-action table, Q-Learning is generally the simplest choice when an off-policy optimal-control target is desired. SARSA is attractive when the behavior policy itself and the cost of exploration must be reflected in learning. DQN becomes appropriate when explicit tabular representation is impossible and the agent must learn useful features from complex observations. Nevertheless, DQN should not be selected merely because it is deeper; its additional computation and instability are justified only when representation capacity is genuinely required.

If implemented, ASE-DQN should be considered successful only if it improves multiple dimensions rather than a single reward score. A convincing outcome would show fewer unsafe actions, lower variance across seeds, improved sample efficiency, and comparable or better final return than standard DQN. If the framework improves safety but significantly reduces performance, or improves reward while increasing instability, the result should be reported as a trade-off rather than as an unconditional improvement. This interpretation approach follows the rubric's requirement for evidence-based and critical analysis.

11. DISCUSSION:

The progression from Q-Learning to SARSA and DQN illustrates a broader theme in artificial intelligence: increasing representational power solves one class of problems while creating new optimization and governance challenges. Tabular methods are transparent but limited in scale. Deep methods are scalable in representation but demand stronger experimental discipline and raise questions about reliability and explainability. This means that future intelligent agents should not be evaluated solely by their ability to maximize reward. They should also be evaluated by how efficiently they learn, how safely they explore, how robustly they generalize, and how clearly their decisions can be inspected.

The literature also suggests that algorithmic improvement should be accompanied by better evaluation practice. Deep-RL research can produce misleading conclusions when only a small number of seeds or selectively reported metrics are used. A technically strong research direction must therefore combine algorithm design with reproducible methodology. The proposed framework deliberately includes repeated-seed testing, ablation analysis, safety metrics, and explanation evaluation so that improvements can be attributed to specific mechanisms rather than to chance.

12. CONCLUSION:

This research review compared Q-Learning, SARSA, and Deep Q-Networks as three important reinforcement-learning approaches for intelligent decision-making. Q-Learning is an off-policy temporal-difference method with strong theoretical foundations in the finite tabular setting. SARSA is an on-policy alternative whose updates incorporate the behavior policy and can therefore respond differently to exploratory risk. DQN extends value-based learning to high-dimensional observations using neural function approximation, experience replay, and a target network, but introduces substantially greater computational and methodological complexity.

The critical review identifies sample efficiency, training stability, value-estimation bias, safe exploration, generalization, interpretability, and reproducibility as continuing research challenges. An Adaptive Safety-Aware Explainable DQN framework is proposed as a feasible direction that combines efficient experience reuse, bias-reduced targets, constrained exploration, and explanation traces.

A reproducible experimental methodology is also defined to validate the proposal using multiple seeds, learning curves, stability measures, efficiency metrics, safety violations, and ablation studies. The main conclusion is that no single reviewed method is universally best; the correct choice depends on the structure of the environment, available computational resources, safety requirements, and the need for scalable representation and generalization.

13.REFERENCES:

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[2] Mnih, V., Kavukcuoglu, K., Silver, D., et al. (2015). Human-level control through deep reinforcement learning. Nature, 518, 529-533. doi:10.1038/nature14236.

[3] Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., & Meger, D. (2018). Deep Reinforcement Learning That Matters. Proceedings of the AAAI Conference on Artificial Intelligence, 32(1). doi:10.1609/aaai.v32i1.11694.

[4] Hessel, M., Modayil, J., van Hasselt, H., Schaul, T., Ostrovski, G., Dabney, W., Horgan, D., Piot, B., Azar, M., & Silver, D. (2018). Rainbow: Combining Improvements in Deep Reinforcement Learning. Proceedings of the AAAI Conference on Artificial Intelligence.

[5] Tang, C., Abbatematteo, B., Hu, J., Chandra, R., Martín-Martín, R., & Stone, P. (2024). Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes. arXiv:2408.03539.

[6] Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.